

Title	Steps to Verify	Expected Result	Status
Verify Email Input Field	1. Open login page. 2. Locate the email input field.	Email input field should be visible and enabled for user input.	Fail
Verify Password Input Field	1. Open login page. 2. Locate the password input field.	Password input field should be visible and masked while typing.	
Verify Login Button Enabled	1. Open login page. 2. Enter valid email and password.	Login button should be enabled after entering valid credentials.	
Verify Error Message on Invalid Login	1. Open login page. 2. Enter invalid email or password. 3. Click Login.	Proper error message should be displayed for invalid credentials.	
Verify Successful Login	1. Open login page. 2. Enter valid email and password. 3. Click Login.	User should be redirected to the dashboard after successful login.	
Verify All Sugar Factories Page Load	1. Open All Sugar Factories page.	Page should load successfully with search bar and two tabs: Company and factory.	
Verify Open Factory Map Button Visibility	1. Login and land on Overview page. 2. Locate "Open factory map" button top-right.	Button should be visible and clickable.	
Verify Open Factory Map Navigation	1. Open Overview page. 2. Click "Open factory map" button.	User should be redirected to the Factories page with a map view showing all factories.	
Verify Total Companies Box	1. Open Overview page. 2. Check "Total companies" box.	Box should display the correct registered company count with label "Registered companies."	
Verify Total Factories Box	1. Open Overview page. 2. Check "Total factories" box.	Box should display the correct monitored factory count with label "Monitored cement factories."	
Verify Cement/Sugar Bags Today Box	1. Open Overview page. 2. Check "bags today" box.	Box should show correct bag count with current date label (e.g., "July 22, 2026").	
Verify Bags This Month Box	1. Open Overview page. 2. Check "bags this month" box.	Box should show correct cumulative bag count with current month/year label (e.g., "July 2026").	
Verify Top Producers Box	1. Open Overview page. 2. Check "Top producers" box.	Box should show "Daily max" count and name, and "This month" count and name, for the top-producing factory.	
Verify Online + Offline Equals Total Factories	1. Open Overview page. 2. Compare "Total factories" value with sum of Online + Offline in connectivity box.	Sum of online and offline should equal total factories count.	
Daily Production & Monthly Production Charts			
Verify Daily Production Chart Defaults to Current Month	1. Open Overview page. 2. Check chart's month/year label below x-axis.	Chart should show data for the current month (e.g., "July, 2026") by default.	
Verify Monthly Production Chart Defaults to Current Year	1. Open Overview page. 2. Check chart's year label below x-axis.	Chart should show data for the current year (e.g., "2026") by default.	
Verify Bar Hover Tooltip	1. Open Overview page. 2. Hover over any bar in Daily/Monthly Production chart.	Tooltip should display exact quantity/date for that bar.	
Verify Clicking a Bar Opens Breakdown Modal	1. Open Overview page. 2. Click on any bar in the Daily or Monthly Production chart.	A "Sugar/Cement Production Breakdown" modal should open with a pie chart and table for that date/month.	
Verify Breakdown Modal Data Accuracy	1. Click a bar to open breakdown modal. 2. Compare table values to actual factory-wise production.	Table should list each factory with correct production count and percentage.	
Verify Breakdown Modal Percentage Sums to 100%	1. Open breakdown modal. 2. Sum all percentage values in the table.	Total should equal 100% (or very close, accounting for rounding).	
Verify Pie Chart Hover Interaction	1. Open breakdown modal. 2. Hover over each pie chart segment.	Tooltip should display corresponding factory name, count, and percentage.	
Verify Pie Chart Segment Colors Match Table	1. Open breakdown modal. 2. Compare pie chart segment colors to color dots in the table.	Each factory's pie segment color should match its corresponding table row indicator.	
Verify Breakdown Modal Close Button	1. Open breakdown modal. 2. Click the "X" close icon.	Modal should close and return to the Overview page.	
Verify Factory Breakdown Tooltip Visibility in Dark Mode	1. Switch the application to dark mode. 2. Open Overview page and scroll to "Factory breakdown" section. 3. Hover over any pie chart segment.	Hover tooltip text/background should have sufficient contrast against the dark theme (not black-on-dark) and should be clearly readable, showing factory name, count, and percentage.	
Factory Status Section			
Verify Factory Status Header Counts	1. Open Overview page. 2. Scroll to "Factory status" section.	Header should show "X online - Y offline" matching the connectivity box counts.	

Verify View All Link	1. Open Factory status section. 2. Click "View all" link.	User should be redirected to the full Factories list page.
Verify Company List Display	1. Open Factory status section. 2. Observe the list below the header.	List should display all companies (e.g., "A.A. Cotton Mills," "Mahmood Textile Mills") with online/offline counts (e.g., "0/1").
Verify Online/Offline Count Format	1. Open Factory status section. 2. Check count format for each company (e.g., "0/1," "0/2").	Format should be "Online count / Total factories" for that company.
Verify Company List Scrollability	1. Open Factory status section. 2. Scroll through the company list.	List should scroll smoothly and load/display all 249 companies (or paginate, depending on implementation).
Verify Expand Icon Rotation	1. Expand "A.A. Cotton Mills". 2. Then click "Mahmood Textile Mills".	Confirm whether only one row can be expanded at a time, or multiple rows can stay open simultaneously — verify against intended design.
Verify Clicking a Factory Opens Factory Page	1. Expand a company row. 2. Click on any factory name in the dropdown.	User should be navigated to that factory's page, which opens by default on the Analytics tab.
Verify Factory Page Side Navigation Tabs	1. Open a factory page.	Left sidebar should show three tabs: "Analytics", "Devices", and "Performance", with the factory name listed above them.
Verify Switching Between Analytics/Devices/Performance Tabs	1. Open a factory page. 2. Click "Devices" tab. 3. Click "Performance" tab. 4. Click "Analytics" tab.	Each tab should load its respective content without errors, and the active tab should be visually highlighted.
Verify Factory Map Button Navigation	1. Open a factory's Analytics page. 2. Click "Factory map" button.	User should be redirected to a page showing the map view for that specific factory's location.
Verify Global Analytics Button Navigation	1. Open a factory's Analytics page. 2. Click "Global analytics" button.	User should be redirected to the main Analytics page showing aggregated data across all factories.
Verify Filters Section Visibility	1. Open a factory's Analytics page. 2. Locate "Filters" section.	Section should be visible with a "Select filter" dropdown and helper text "Changes apply automatically when a valid period is selected."
Verify By Date Filter	1. Select "By date" filter. 2. Choose a specific date.	Data below (bag counts, hourly graph) should update automatically to reflect the selected date, without needing a manual refresh/apply action.
Verify By Month Filter	1. Select "By month" filter. 2. Choose a specific month.	Data below should update automatically to reflect totals for the selected month.
Verify Date Range Filter - Valid Range	1. Select "Date range" filter. 2. Choose a start date and an end date after the start date.	Data below should update to reflect totals for the selected date range.
Verify Date Range Filter - End Date Before Start Date	1. Select "Date range" filter. 2. Choose a start date. 3. Attempt to choose an end date earlier than the start date.	System should prevent selection or show a validation error; end date must be after the start date.
Verify Cement Bags (Day) Box	1. Open a factory's Analytics page. 2. Check "Cement bags (day)" box.	Box should display correct bag count aggregated across all packer lines for the current/selected day, with date label below.
Verify Cement Bags (Month) Box	1. Open a factory's Analytics page. 2. Check "Cement bags (month)" box.	Box should display correct bag count aggregated across all packer lines for the current/selected month, with month/year label below.
Verify Totals Boxes Update with Filter Change	1. Apply a date/month/range filter. 2. Observe the two totals boxes.	Both boxes should update to reflect totals for the newly selected filter period.
Verify Packer-Line-Specific Bag Counts Match Filter	1. Select a packer line. 2. Apply a date filter.	Cement bags (day) and "Cement bags (month)" boxes under the packer line should reflect data specific to that device and the selected filter.
Verify Hourly Graph with Single-Day (By Date) Filter	1. Apply "By date" filter for a specific single day. 2. Check "Hourly date" field and graph.	Hourly graph should display hour-by-hour bag count data for that exact selected date.
Verify Hourly Graph with Current Month Filter	1. Apply "By month" filter for the current month.	Hourly graph should default to showing the current day's hourly data (not the 1st of the month).
Verify Hourly Graph with Past Month Filter	1. Apply "By month" filter and select a previous month (not current).	Hourly graph should default to showing the 1st day of that selected month's hourly data.
Verify Hourly Graph with Date Range Filter	1. Apply "Date range" filter with a valid start and end date.	Hourly graph should default to showing hourly data for the first (start) date of the selected range.
Verify Live View Tab Selection	1. Select a packer line. 2. Click "Live view" tab.	Tab should switch to Live View, showing "Live view · [Device Name]" with quality, frame rate, and video preview panel.

Verify Live View Description Text	1. Open Live view tab.	Helper text should read "On-demand annotated stream — complete-but-delayed, not recorded."
Verify Quality/Resolution Dropdown Options	1. Open Live view tab. 2. Click "Quality / resolution" dropdown.	Dropdown should list resolution options: 360p, 720p, and 1080p, each with an approximate Mbps bitrate label.
Verify Frame Rate Dropdown Options	1. Open Live view tab. 2. Click "Frame rate" dropdown.	Dropdown should list frame rate options: 1 fps, 2 fps, 5 fps, and 10 fps.
Verify Idle State Before Go Live	1. Open Live view tab without clicking "Go live".	Video panel should show status "idle" and overlay text "idle — press Go Live".
Verify Devices Table Columns	1. Open Devices page.	Table should display columns: S.NO, DEVICE NAME, DEVICE TYPE, STATUS, CREATED ON, UPDATED ON, ACTIONS.
Verify Search Devices Field	1. Open Devices page. 2. Type a device name in the search box.	Table should filter and show only devices matching the search term.
Verify Edit Icon Action	1. Open Devices page. 2. Click the pencil (edit) icon for a device.	Edit form/modal should open pre-filled with that device's current name and type.
Verify Device Deletion Confirmation	1. Click delete icon. 2. Confirm deletion in the prompt.	Device should be removed from the table and the list should refresh.
Verify One-Day Filter Visibility	1. Open Performance page.	
Verify Selecting a Specific Day	1. Open Performance page. 2. Select a specific date using the filter.	A date/day filter control should be visible, allowing selection of a specific day. GPU usage, CPU usage, and internet speed data should update to reflect that selected day only.
Verify Specific Time-of-Day Data Accuracy	1. Select a day filter. 2. Check a specific hour's data point (e.g., 12:00 PM) on the graph.	Data point should accurately reflect the actual recorded usage/speed at that specific time.
Verify Download Functionality	1. Apply a day filter. 2. Click the download button.	A file (e.g., PDF/CSV) should be downloaded containing performance data for the selected day.
Verify Downloaded File Matches Selected Time Range	1. Apply a day filter (e.g., 12:00 AM to 12:00 AM of a specific day). 2. Download the report. 3. Open the downloaded file.	Downloaded file's data should exactly match the time range selected on screen (12 AM–12 AM of that day), with no extra or missing data.
Verify Download Without Selecting a Filter	1. Open Performance page without applying any filter. 2. Click download.	File should download using the default (current day) time range.

Factory Edit Page

Verify Update Factory Modal Opens	1. On Factories page, click edit icon for a factory.	Modal should open titled "Update Factory" with subtext "Update the factory details."
Verify Close (X) Icon	1. Open modal. 2. Click "X" icon top-right.	Modal should close without saving any changes.
Verify Cancel Button	1. Open modal. 2. Modify a field. 3. Click "Cancel."	Modal should close and discard all unsaved changes.
Verify Modal Scroll Behavior	1. Open modal. 2. Scroll down.	All fields (down to Factory Images and Update button) should be reachable via scroll, with header/scrollbar behaving correctly.
Verify Factory Name - Valid Update	1. Clear "Factory Name" field. 2. Enter a valid new name. 3. Click "Update."	Factory name should update successfully and reflect on the Factories list.
Verify Factory Name - Special Characters	1. Enter name with special characters (e.g., "@Test#Factory!"). 2. Click "Update."	Confirm whether special characters are accepted or rejected per validation rules.
Verify Factory Name - Exceeds Max Length (Negative)	1. Enter a name exceeding the allowed character limit. 2. Click "Update."	Validation error should appear, or input should be restricted from exceeding max length.
Verify Parent Company - Valid Selection	1. Select a different parent company. 2. Click "Update."	Factory should be reassigned to the newly selected company.
Verify GPS Latitude - Out of Range (Negative)	1. Enter a latitude outside -90 to 90 (e.g., 200). 2. Click "Update."	Validation error should appear; update should be blocked.
Verify GPS Longitude - Out of Range (Negative)	1. Enter a longitude outside -180 to 180 (e.g., 400). 2. Click "Update."	Validation error should appear; update should be blocked.
Verify Ownership Structure - Valid Update	1. Update text in "Ownership Structure" field. 2. Click "Update."	Value should save successfully.
Verify Company Website - Valid URL	1. Enter a valid URL (e.g., https://www.example.com). 2. Click "Update."	URL should save successfully.

Verify Company Website - Invalid URL Format (Negative)	1. Enter an invalid URL (e.g., "wwwexamplecom" or "http://test"). 2. Click "Update."	Validation error should appear indicating invalid URL format.
Verify Company Website - Empty Field	1. Clear the website field. 2. Click "Update."	Confirm if field is required or optional; behave accordingly.
Verify Aggregation Start Hour Dropdown Options	1. Click "Aggregation day-window start hour" dropdown.	Dropdown should list hour options (e.g., 00:00–23:00).
Verify Aggregation Start Hour Default Value	1. Open modal for a factory that hasn't customized this field.	Field should default to "06:00" as stated in the helper text.
Verify Aggregation Start Hour Change Does Not Affect Past Data	1. Change start hour and save. 2. Check previously recorded analytics/production data.	Past aggregated data should remain unchanged; only future aggregations should use the new hour.
Verify Production Capacity - Valid Numeric Value	1. Enter a valid number (e.g., 5000). 2. Click "Update."	Value should save successfully.
Verify Production Capacity - Negative Number (Negative)	1. Enter a negative value (e.g., -100). 2. Click "Update."	Validation error should appear; negative values should be rejected.
Verify Production Capacity - Non-Numeric Input (Negative)	1. Enter letters (e.g., "abcd"). 2. Click "Update."	Field should reject non-numeric input or show validation error.
Verify Production Capacity - Decimal Values	1. Enter a decimal (e.g., 5000.5). 2. Click "Update."	Confirm whether decimals are accepted or restricted to whole numbers.
Verify Unit Field - Valid Update	1. Update "Unit" field text (e.g., "bags/annum" to "tons/month"). 2. Click "Update."	Value should save successfully.
Verify Public Company Checkbox - Checked State	1. Check "Public company (listed)" checkbox. 2. Click "Update."	Value should save as checked/true.
Verify Public Company Checkbox - Unchecked State	1. Leave checkbox unchecked (default). 2. Click "Update."	Value should save as unchecked/false.
Verify Company Logo Display	. Open modal for a factory with an existing logo.	Current logo thumbnail should display correctly.
Verify Company Logo - Change (Valid File)	1. Click "Change." 2. Upload a valid PNG/JPEG/SVG under 2MB.	New logo should replace the old one and preview should update.
Verify Company Logo - Remove	1. Click "Remove" on existing logo. 2. Click "Update."	Logo should be removed and factory should show no logo/default placeholder.
Verify Add Images - Valid Upload	. Click "Add images." 2. Upload one or more valid image files (within the 10-image limit).	Images should upload and appear as new thumbnails with caption fields.
Verify Add Images - Exceeds 10-Image Limit (Negative)	1. Attempt to add images beyond the 10-image cap.	System should block additional uploads and show a limit-reached message.
Verify Remove Image	1. Click the "X" icon on an existing factory image.	Image should be removed from the list immediately (pending save).
Verify Update Button - All Valid Fields	1. Fill/edit all fields with valid data. 2. Click "Update."	Form should submit successfully, modal should close, and Factories list should reflect updated data.
Verify Update Button - No Changes Made	1. Open modal. 2. Without changing anything, click "Update."	Form should submit successfully with unchanged data
Verify Update Button Disabled State During Submission	1. Fill valid data. 2. Click "Update."	Button should show a loading state and be disabled to prevent duplicate submissions while processing.

ANALYTICS

Verify Analytics Page Load	1. Open Main Analytics page.	Page should load successfully with factory filter, date filter, and analytics cards/graphs.
Verify Default Data on First Load	1. Open page without applying any filter.	Daily and monthly production cards, daily & monthly graphs, and line chart should show current day data .
Verify Factory Filter Visibility	1. Open page.	Factory filter dropdown should be visible and enabled.
Verify Date Filter Visibility	1. Open page.	Filter by date dropdown should be visible with options: By Date, By Month .
Verify Factory Selection	1. Select any factory from factory filter dropdown.	Selected factory should be applied successfully.

Verify Daily & Monthly Cards with Factory Selected	1. Select factory without selecting date filter.	Daily and monthly production cards should update to show current day data for selected factory .
Verify By Date Filter Functionality	1. Select factory. 2. Apply By Date filter. 3. Choose a specific date.	Daily & monthly cards, daily production graph, monthly production graph, and line chart should show specific date data .
Verify By Month Filter Functionality	1. Select factory. 2. Apply By Month filter. 3. Choose a month.	All cards and graphs should show full month data for the selected factory.
Verify Daily Production Graph Update	1. Apply By Date filter.	Daily production graph should display data only for selected date.
Verify Monthly Production Graph Update	1. Apply By Date filter.	Monthly production graph should display data only for the month of selected date .
Verify Line Chart Update By Date	1. Apply By Date filter.	Line chart should update to reflect production data of the selected date.
Verify Daily Production Graph By Month	1. Apply By Month filter.	Daily production graph should display data of all days of the selected month .
Verify Monthly Production Graph By Month	1. Apply By Month filter.	Monthly production graph should display month summary data .
Verify Line Chart By Month	1. Apply By Month filter.	Line chart should display data for all days of selected month .
Verify Reset / Clear Filter	1. Apply any filter. 2. Click reset/clear button.	Page should return to default current day data .
Verify Changing Factory Updates Data	1. Select factory A. 2. Select factory B.	All cards and graphs should update according to factory B data.
Verify Switching Filter Type	1. Apply By Date filter. 2. Switch to By Month filter.	Cards and graphs should update according to the new filter type.
Verify Invalid Date Handling	1. Select By Date filter. 2. Choose invalid date or future date.	Proper validation or “No Data Available” message should display.
Verify API Data Integration	1. Apply factory and date/month filters.	All cards and graphs should display data retrieved from API.
Verify No Factory Selected Behavior	1. Open filter and do not select a factory.	Page should show default analytics data or prompt user to select a factory.
Verify UI Responsiveness	1. Open analytics page on mobile, tablet, and desktop.	Cards, graphs, and filters should display correctly on all screen sizes.
Verify Graph Synchronization with Cards	1. Apply any filter.	Daily and monthly production cards should be consistent with daily/monthly graphs and line chart.
Verify Edge Case – Start/End Month	1. Apply By Month filter selecting month boundaries.	Graphs and cards should accurately reflect production data for first and last day of month.
Verify Combined Date Range and Factory	1. Select factory and apply By Date filter.	Cards and graphs should reflect both selected factory and selected date .
Verify Line Chart Tooltips	1. Hover over line chart points.	Tooltip should display correct production value and timestamp.
Verify Tooltip on Bar Graph	1. Open Analytics page. 2. Hover over any bar in the bar graph.	Tooltip should appear showing the correct production value and timestamp.
Verify Tooltip on Line Chart	1. Open Analytics page. 2. Hover over any point on the line chart.	Tooltip should appear showing correct production value and timestamp.
Verify Click on Tooltip Opens Pie Chart	1. Hover over a bar/point. 2. Click on the tooltip.	Pie chart modal should open displaying factory breakdown data.
Verify Pie Chart Data Accuracy	1. Open pie chart from tooltip. 2. Compare values with backend or expected data.	Pie chart should accurately reflect the factory breakdown data.
Verify Pie Chart Close Functionality	1. Open pie chart from tooltip. 2. Click Close button or outside modal.	Pie chart modal should close successfully.
Verify Pie Chart Responsiveness	1. Open pie chart on mobile, tablet, and desktop.	Pie chart should display correctly on all screen sizes without distortion.
Verify Pie Chart for Different Filters	1. Apply different date or month filters. 2. Click tooltip to open pie chart.	Pie chart should update according to the selected filter data.

REPORTS

Verify Report Page Load	1. Open Report page.	Report page should load successfully with filters and empty results table.
Verify Fiscal Year Dropdown	1. Open Report page.	Fiscal year dropdown should display available options (e.g., 2023–2024, 2024–2025).

Verify Buttons Visibility After Fiscal Year Selection	1. Select a fiscal year.	Apply Filter and Download buttons should become visible/enabled.
Verify Apply Filter Without Fiscal Year	1. Open Report page. 2. Do not select fiscal year.	Apply Filter button should remain disabled or show validation message.
Verify Fiscal Year Data Range	1. Select fiscal year 2024–2025. 2. Click Apply Filter.	Table should show data from July 2024 to June 2025 .
Verify Quarter Filter Availability	1. Select a fiscal year.	Quarter filter (Q1–Q4) should be visible and selectable.
Verify First Quarter Data	1. Select fiscal year. 2. Select Q1. 3. Apply filter.	Table should show data for July, August, September only.
Verify Second Quarter Data	1. Select fiscal year. 2. Select Q2. 3. Apply filter.	Table should show data for October, November, December only.
Verify Third Quarter Data	1. Select fiscal year. 2. Select Q3. 3. Apply filter.	Table should show data for January, February, March only.
Verify Fourth Quarter Data	1. Select fiscal year. 2. Select Q4. 3. Apply filter.	Table should show data for April, May, June only.
Verify Month Selection	1. Select fiscal year. 2. Select a specific month. 3. Apply filter.	Table should show data only for the selected month.
Verify Specific Date Selection	1. Select fiscal year. 2. Select a specific date. 3. Apply filter.	Table should show data only for the selected date.
Verify Company Filter	1. Select fiscal year. 2. Select a specific company. 3. Apply filter.	Table should display data only for selected company.
Verify Multiple Companies Selection	1. Select multiple companies. 2. Apply filter.	Table should show combined data for selected companies.
Verify Factory Filter	1. Select a company. 2. Select one or more factories. 3. Apply filter.	Table should show data for selected factories only.
Verify All Factories Selection	1. Select “All Factories”. 2. Apply filter.	Table should show data for all factories under selected company/fiscal year.
Verify Packer Filter	1. Select a factory. 2. Select a specific packer. 3. Apply filter.	Table should show data only for selected packer.
Verify Combined Filters	1. Select fiscal year + quarter/month + company + factory + packer. 2. Apply filter.	Table should reflect data according to all selected filters.
Verify Table Data Accuracy	1. Apply filters. 2. Verify totals and rows.	Table data should match backend results and applied filters.
Verify No Data Scenario	1. Apply filters with no available data.	Table should show “No data available” message.
Verify Download PDF	1. Apply filters. 2. Click Download PDF.	PDF file should download with correct filtered data.
Verify Download Excel	1. Apply filters. 2. Click Download Excel.	Excel file should download with correct filtered data.
Verify Download Without Apply Filter	1. Select fiscal year but do not click Apply Filter. 2. Click Download.	System should prompt to apply filter first or auto-apply filters.
Verify Download Data Consistency	1. Apply filters. 2. Compare table data with downloaded file.	Downloaded PDF/Excel data should exactly match table data.
Verify Reset Filters	1. Apply multiple filters. 2. Click Reset/Clear.	All filters should reset and table should clear or return to default state.
Verify Pagination on Report Table	1. Apply filters with large dataset.	Table pagination should work correctly.
Verify Performance	1. Apply filters with large data.	Filters and downloads should complete within acceptable time.

Contact List

Verify Contact List Page Loads	1. Navigate to the Contact List page.	Page should load with the title, search bar, Add contact button, and contact table.
Verify Table Columns	1. Open the Contact List page.	Table should show columns: S.No, Name, Email, Address, Number, Factory, Actions.
Verify Existing Contacts Displayed	1. Open the page with existing contacts.	All saved contacts should be listed with correct details under each column.
Verify Empty State	1. Open the page with no contacts.	An appropriate empty/"No contacts" state should be displayed.
Verify Serial Numbering	1. View multiple contacts.	S.No should increment correctly and sequentially.
Verify Add Contact Modal Opens	1. Click "Add contact".	The Add Contact modal should open with all fields and Cancel/Submit buttons.
Verify All Form Fields Present	1. Open the modal.	Fields should include Name, Email, Address, Contact number, and Factory dropdown.
Verify Factory Dropdown Options	1. Open the Factory dropdown.	Available factories should be listed and selectable.
Verify Add Contact With Valid Data	1. Fill all fields with valid data. 2. Select a factory. 3. Click Submit.	Contact should be created and appear in the table with correct details.
Verify Contact Linked to Selected Factory	1. Create a contact with a specific factory.	1. Create a contact with a specific factory.
Verify Cancel Button	1. Open modal. 2. Enter data. 3. Click Cancel.	Modal should close without saving; no contact should be added.

Verify Close (X) Button	1. Open modal. 2. Click the X icon.
Verify Form Resets After Submit	. Add a contact. 2. Reopen the modal.
Verify Success Feedback	1. Submit a valid contact.
Verify Submit With All Fields Empty	1. Click Submit without entering anything.
Verify Name Required	1. Leave Name empty. 2. Fill others. 3. Submit.
Verify Email Required	1. Leave Email empty. 2. Submit.
Verify Invalid Email Format	1. Enter "abc@" or "abc.com". 2. Submit.
Verify Factory Required	1. Leave Factory unselected. 2. Submit.
Verify Phone Number Length	Length1. Enter a too-short or too-long number.
Verify Search by Name	1. Type a contact's name in Search.
Verify Search by Email	1. Search by an email value.
Verify Search by Factory	1. Search by a factory name.
Verify Clear Search	1. Enter a search term. 2. Clear it.
Verify Edit Contact	1. Click Edit on a contact. 2. Change details. 3. Save.
Verify Edit Validation	1. Edit a contact with invalid data. 2. Save.
Verify Change Assigned Factory	1. Edit a contact. 2. Select a different factory. 3. Save.
Verify Cancel Edit	1. Open Edit. 2. Change data. 3. Cancel.
Verify Delete Contact	1. Click Delete on a contact. 2. Confirm.
Verify Delete Confirmation	1. Click Delete.
Verify Cancel Delete	1. Click Delete. 2. Cancel the confirmation.
Verify S.No Reorders After Delete	1. Delete a contact.
Verify Pagination (if many contacts)	1. Add enough contacts to exceed one page.

Modal should close without saving.
Fields should be empty (no leftover data from the previous entry).
A success message/confirmation should be displayed.
Validation messages should appear for required fields; contact should not be saved.
Validation should require Name.
Validation should require Email (if mandatory).
Validation should reject the invalid email format.
Validation should require a factory to be selected.
Validation should enforce the allowed phone-number length.
Only matching contacts should be displayed.
Matching contacts should be displayed.
Contacts linked to that factory should be displayed.
Full contact list should be restored.
Contact should update with the new details in the table.
Same validation rules as Add should apply.
The contact's factory association should update accordingly.
Changes should be discarded; original data retained.
Contact should be removed from the list.
A confirmation prompt should appear before deletion.
Contact should remain in the list.
Remaining contacts' S.No should update sequentially.
Pagination should work and display contacts correctly.

NOTIFICATIONS

Verify Notification Page Load	1. Open Notification page.
Verify User Notification for Company Creation	1. Create a new company.
Verify User Notification for Company Update	1. Update a company record.
Verify User Notification for Company Deletion	1. Delete a company.
Verify User Notification for Factory Creation/Update/Delete	1. Create, update, or delete a factory.
Verify User Notification for Device Creation/Update/Delete	1. Create, update, or delete a device.
Verify System Notification for Camera Offline	1. Simulate camera offline for 15 minutes.
Verify System Notification for Belt Running with No Production	1. Simulate belt running but production count not increasing.
Verify System Notification for Network Error	1. Simulate network error.
Verify Notification Details	1. Click on any notification.
Verify Notification Type Filtering	1. Filter notifications by User Notification or System Notification.
Verify Notification Mark as Read	1. Open notification. 2. Mark it as read.
Verify Notification Count	1. Generate multiple notifications.

Page should load successfully with tabs/sections for **User Notification** and **System Notification**.

A user notification should appear indicating the company was created successfully.

A user notification should appear indicating the company was updated successfully.

A user notification should appear indicating the company was deleted successfully.

User notification should appear for each action, showing the correct message.

User notification should appear for each action with correct details.

System notification should appear indicating the camera is offline.

System notification should appear indicating production issue.

System notification should appear indicating network failure.

Notification details should display full information including timestamp and affected entity.

Only selected type of notifications should be displayed.

Notification should change status to read and may update UI (greyed out or removed from unread count).

Notification count badge should update correctly.

Verify Notification Sorting	1. Check notifications order.	Notifications should be sorted by timestamp descending (latest first).
Verify Notification Pagination / Load More	1. Scroll to bottom or click Load More.	Additional notifications should load without errors.
Verify Notification UI Responsiveness	1. Open notification page on mobile, tablet, desktop.	UI should be responsive and all notifications visible properly.
Verify Real-time Notification	1. Trigger user or system notification while page is open.	Notification should appear in real-time without page refresh.
Verify Deletion of Notification	1. Delete a notification (if delete option available).	Notification should be removed successfully.
Verify No Notification State	1. Clear all notifications or open page with no notifications.	Proper “No Notifications Available” message should display.
Verify Notification Persistence	1. Generate notification. 2. Refresh page.	Notification should persist after page refresh.
Verify Default Pagination Limit	1. Open Notification page.	By default, 10 notifications should be displayed per page.
Verify Change Pagination Limit	1. Open Notification page. 2. Change pagination dropdown to 25, 50, or 100 .	Notifications displayed per page should update according to selected value.
Verify Pagination Navigation	1. Open Notification page with more notifications than current page limit. 2. Click Next / Previous page or page numbers.	Notifications should navigate correctly to the next/previous page without errors.